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GeneNet: Transfer learning-based hybrid African buffalo optimization with genetic algorithm for gene expression based cancer classification

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ABSTRACT

Gene therapy is one of the advanced medical diagnosis environments, which is used to identify the solutions to various cancers by identifying the perfect gene expressions. The computer aided detection of gene expressions is a research problem, where cancer type is identified by artificial intelligence methods. However, the conventional machine learning models were failed to provide the accurate cancer classification and resulted in poor quantitative performance. Therefore, this work focuses on the implementation of gene expression-based cancer classification (GECC) network, here after referred as GeneNet. Initially, the proposed GeneNet performs the data preprocessing to normalize the gene-expression data by replacing the unknown characters and missing symbols. Then, the visual geometry group developed VGG16 model for extraction of correlated, and structural features from pre-processed dataset. In addition, hybrid African buffalo optimization with genetic algorithm (HABO-GA) is used to select the optimal features through natural inspired hunting (searching) properties. Finally, low complex convolutional neural network (LC-CNN) model is trained with HABO-GA features, which also performs the multi-class cancer classification from test data. The simulations are conducted on publicly available GECC datasets such as lymphography, ovarian, leukemia, lung, and prostate cancers discloses the superiority of proposed GeneNetas compared to state-of-the-art approaches with 86.6 %, 100 %, 76.92 %, 100 %, and 100 % of accuracy, respectively.

Introduction

Microarray technology is a leading-edge technology in molecular biology when it comes to the contribution of information in quantitating hundreds or thousands of genes that are used in the diagnosis of different diseases and to predict the possible outcomes of an ailment [1]. The genes which are regulated due to disease condition can be analysed through the expression extracted from the microarray data samples. These measurements also assist in the investigation of cancer for clinical medicine at the biology and molecular level [2]. Cancer can alter the gene expression profile of the body cells. This fatal genetic disease transpired from the mutations or epigenetic changes. Therefore, the microarray data is utilized in clinical diagnosis to recognize down or up the regulated gene expression [3], which is the reason for activating some oncogenic pathways, generating new biomarkers, and leading to cancer disease. Nevertheless, this approach comes with a considerable cost and time. Moreover, it is not clinically applicable to all patients [4]. The algorithms used in the data analysis are not helping the researchers either due to their restrictions, which is a massive setback for microarray technology. Microarray array data analysis has been used as a source for profiling gene expression for many decades [5]. However, it suffers from noise and the difficulty of range detection as it involves transcriptome and genome references. Mainly, it utilizes the sequence-specific probe hybridization accompanied by fluorescence detection to estimate gene expression levels. The genes that play a crucial role while specifying the phenotype is recognized when compared over the profiles of gene expression from various types of tissues. Several kinds of clinical courses are required for the classification and prediction of cancer. Also, the diagnosis of cancer is very slow. The machine learning (ML) [6] is devised to overcome the problems in the conventional methods.

Machine learning is a branch of artificial intelligence which is used to identify associations among data by finding underlying patterns using past experience and learning. ML is becoming indispensable in the age of mass data, given that it is becoming increasingly difficult for humans to find trends and patterns in data to predict future outcomes [7]. Hence, machine learning is replacing humans to identify underlying patterns in

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data and make predictions for the future to make proper decisions. ML extracts features itself with almost zero human intervention, and then uses these features to make predictions. ML is being implemented almost everywhere. Its typical applications [8] are in natural language processing, forecasting, flight governance, and biology to detect sequences of proteins and RNA. The effectiveness of genes selection is assessed by the accuracy of classification methods, which is highly crucial. There is also a variety of machine learning-based classification methods that can be used with genes feature selection for improving classification accuracy results.

Usually, feature selection [9] is used for the selection of significant information for the considered problems. The different methods used for it are wrapper based: ensemble-based, filter-based, and embedded or hybrid-based models. In order to select the feature subsets freely of any learning method, the filter employs a threshold value and the relevance score. The wrapper models employ the predictive precision of foreordained learning techniques. The process of the embedded model permits various classes of learning model interaction. In addition, several meta-heuristic-based optimized algorithms such as grey-wolf optimization, whale optimization, deer hunting, and red deer optimization etc. have been introduced in recent years due to their broad applicability, ease of implementation, hybridization, efficiency, and flexibility as compared to heuristic optimization approaches. The artificial immune model operates by imitating the exempt response, memory, and regulation model of the biological immune model by attaining the best performance, strength, and self-association. Besides, the diagnosis of any human disease is a very challenging task for the people who are involved in the healthcare organization by enhancing the accuracy of disease classification is essential with selection of appropriate features. Therefore, the principal contributions of this work are as follows:

- Development of transfer learning -based deep feature extraction from pre-processed dataset using VGG16 model.
- Adopting naturally inspired meta-heuristic optimization approach named HABO-GA for selecting the best features with high correlation properties, where GA is used to optimize the weights of HABO.
- Utilizing the LC-CNN for classification of multiple cancers from test gene-expression dataset.

The rest of the article is organized as follows: Section 2 covered the literature survey with problem analysis. Section 3 covers the detailed analysis of proposed GeneNet methodology. Section 4 describes the results and discussion of proposed GeneNet and its performance comparison with state-of-the-art methods using multiple datasets. Finally, Section 5 explains the conclusion with future scope followed by bibliography.

Related work

This section discusses the related works of GECC approaches with their shortcomings. In [10], Dwivedi and Ashok Kumar adopted artificial neural network for gene expression classification, which is a tenfold cross-validation. Furthermore, all samples were successfully identified, and models were confirmed using independent test data. However, this work suffered from overfitting, and higher computational complexity. Liu et al. [11] presented a versatile strategy for categorising cancer gene expression by feature gene selection and adjusting parameters while utilising various datasets via cross validation. Six conventional approaches were used to compare the performance of the suggested method, which was shown to be superior in terms of finding feature cancer genes. However, it is difficult to choose a right kernel and it also suffered from incomprehensibility. Ayyad et al. [12] proposed modified k-nearest neighbour (MKNN), a novel classification method for gene expression data, is used in the smallest modified KNN (SMKNN) and biggest modified KNN situations (LMKNN). These two implementations are designed to improve KNN performance. Utilizing strong neighbours from training data while utilising a novel weighting approach is the main concept. However, the feature selection approach hasn't addressed and it's very difficult with the KNN to extract the deep features. In [13–15], authors examined and contrasted the most modern hybrid strategies that use evolutionary techniques wrapped with biological inspiration. In addition, they have also presented various new approaches for GECC with their shortcomings and possible future solutions to enhance the classification accuracy. In [16], Pati proposed lung cancer gene expression data may be found in the Kent ridge bio-medical dataset repository. To identify and forecast the best group of genes that are the most likely cause of lung cancer, the microarray gene expression data are examined. But this work tested with only lung cancer dataset. In addition, it employed only machine learning algorithms for cancer classification which produced lesser accuracy as compared to deep learning models. In [17], Xu et al. proposed dimensionality reduction of gene expression data using the fisher ratio and a local rough set to improve classification performance. The tests on the RNA-seq gene expression data shown that their gene selection technique has greater accuracy with fewer genes and the suggested deep flexible neural forest (DFNForest) model has superior performance for categorization of cancer subtypes as compared to the standard methods. However, they have employed particle swarm optimization for parameter optimization which has low convergence rate in the process of iteration. In [18] authors proposed novel CNN architecture for RNA-Seq gene expression data-based breast cancer categorization, where *R* studio is used to get the gene expression data from Pan-Cancer Atlas and the "Illumina HiSeq" platform is utilised to download the data. The pre-processing involves deleting outlier samples from the downloaded data, which are identified using the array-array intensity correlation, which creates a symmetric square matrix of Spearman correlation between samples. However, the features selection hasn't addressed, and they have tested only breast cancer gene data.

Wang et al. proposed to choose cancer genes in groups with the usage of weighted general group lasso (WGGL) model [19]. Based on a weighted gene co-expression network analysis, a gene grouping heuristic technique is provided. Khorshed et al. proposed a multi-tissue cancer classifier based on whole-transcriptome gene expressions and presented a deep learning framework for cancer detection [20]. They have presented a brand-new CNN architecture that is especially made to handle the complexity of whole-transcriptome gene expressions and shown how it can be applied as a general end-to-end learning system for classification across various cancer tissue types without undergoing the necessary step of gene feature selection. However, still it is required to improve the classification accuracy with the design of ensemble models that can integrate numerous genome platforms. Diaz et al. proposed genetic algorithm for grouping, which can solve a problem of maximum group diversity [21]. It has been used to categorise an uneven collection of 801 samples of RNA-Seq. data on gene expression in 5 different forms of cancer from 20,531 genes in each sample. It also extracts several gene groupings with good multiple classification accuracy. However, it suffered with incomplete exploration of the solution space. The gene expression patterns of the two main subtypes of lung cancer, lung AC and lung SCC has been examined by Yuan et al. [22]. They have employed several machines learning techniques, including rule learning algorithms, traditional classification algorithms, and feature selection approaches. In addition, the have also discovered and examined several potential differentially expressed genes and expression patterns. However, this approach addressed only lung cancer related issues.

In [23], the gene expression patterns for the categorization of supplied cancer data, a hybrid deep learning model based on Laplacian score-convolutional neural network (LS-CNN) is examined by Shah et al., where 10 distinct benchmark datasets are tested and the performance of system was assessed using a variety of performance assessment criteria, including accuracy and confusion matrix. In [24], the TLBOGSA approach is implemented based on a novel hybrid wrapper strategy that combines the traits of gravitational search algorithm with instructional

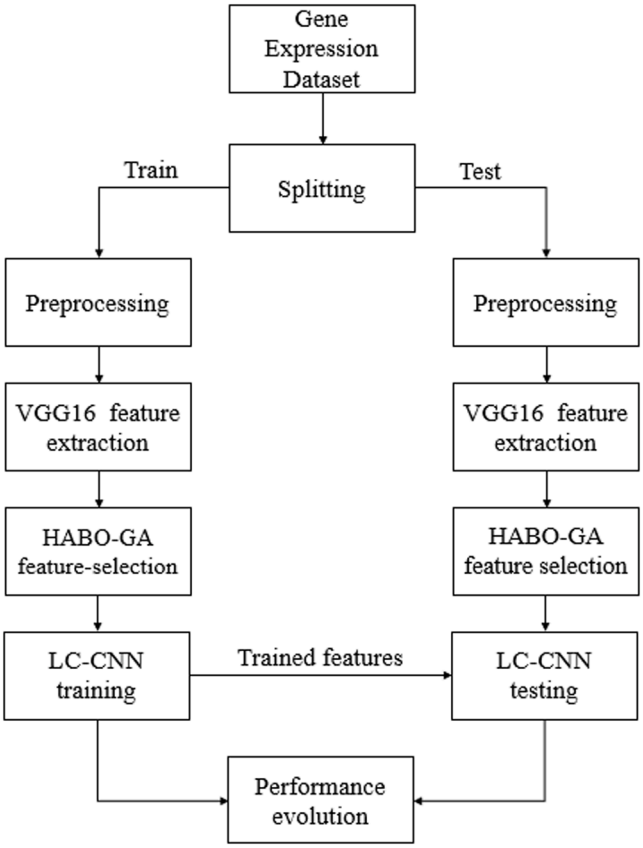


Fig. 1. Block diagram of proposed GeneNet.

learning-based algorithm (TLBO). To convert the continuous search space to binary search space and create binary TLBOGSA, a new encoding technique is also included into TLBOGSA. However, it relies on an often-faulty assumption of equally important and independent features which results in biased posterior probabilities. In [25], authors addressed an innovative approach for classifying cancer based on the Jaya Ant lion optimization-based deep recurrent neural network, which assumed various stages such as data normalisation, data transformation, feature dimension detection, and classification. But it has a long runtime at the end of the optimization process. Houssein et al. proposed the hybrid approach, which combines the Barnacles mating optimizer (BMO) and support vector machine (SVM) algorithms for microarray gene expression profiling to identify the most accurate and useful genes for cancer categorization [26].However, this work needs be extended with neural networks for enhanced classification accuracy.

Koul and Manvi [27] presented a system for categorising cancer gene expression patterns. The framework is made up of a series of techniques for feature selection, classification, and data pre-processing, where standard feature scaling and normalisation are utilized for data pre-processing. They have employed recursive feature elimination and genetic algorithm for feature selection and finally deep neural networks with hyperparameter optimization are used for classification. However, this work utilized meta-pool of diverse classifiers that resulted in higher computational complexity with classification uncertainty. In [28], a deep neural network-based lung cancer prediction model with KL divergence gene selection is implemented to address the issues of high dimensionality, a lack of samples, and model unfitting on gene expression data. In addition, it performed better on the verification data set than standard algorithms and it also has the benefits of quick training and high accuracy. In [29], authors implemented the GECC-Net with AlexNet based feature extraction, hybrid fuzzy ranking network (HFRN) based feature selection, and multi kernel-SVM based classification. The advantage of this work is that they introduced HFRN based optimal feature selection process for improving performance. To this end, various related work on GECC approaches is presented and addressed their limitations which includes the optimal feature selection, and transfer learning-based feature extraction for improved classification performance.

Table 1
VGG16 layer sizes.

	Layer	Feature map	Feature size	Kernel size	Stride	Activation
1	2 X Convolution	64	224 × 224 × 64	3 × 3	1	ReLU
	Max pooling	64	112 × 112 × 64	3 × 3	2	ReLU
3	2 X Convolution	128	112 × 112 × 128	3 × 3	1	ReLU
	Max pooling	128	112 × 112 × 128	3 × 3	2	ReLU
5	2 X Convolution	256	112 × 112 × 128	3 × 3	1	ReLU
	Max pooling	256	28 × 28 × 512	3 × 3	2	ReLU
7	3 X Convolution	512	28 × 28 × 512	3 × 3	1	ReLU
	Max pooling	512	14 × 14 × 512	3 × 3	2	ReLU
10	2 X Convolution	512	14 × 14 × 512	3 × 3	1	ReLU
	Max pooling	512	7 × 7 × 512	3 × 3	2	ReLU
13	FC	–	25,088	3 × 3	–	ReLU
14	FC	–	4096	3 × 3	–	ReLU
15	FC	–	4096	3 × 3	–	ReLU
Output	FC	–	1000	3 × 3	–	SoftMax

Proposed methodology

This section gives the detailed analysis of proposed GeneNet. Fig. 1 shows the operational diagram of GeneNet, which is used to classify the disease from lymphography, ovarian, leukemia, lung, and prostate cancer-based gene expression datasets. As shown in Fig. 1, there are four phases in proposed GeneNet which includes preprocessing, feature extraction, feature selection and classification. First, the dataset

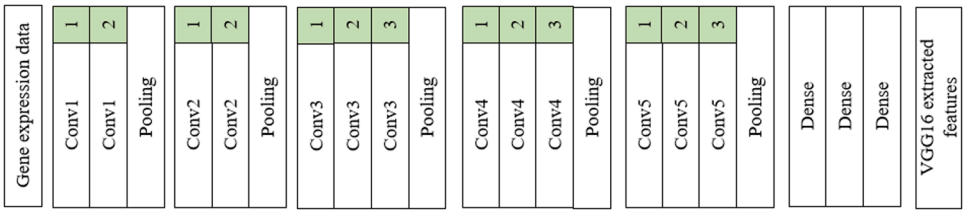


Fig. 2. VGG16 architecture.

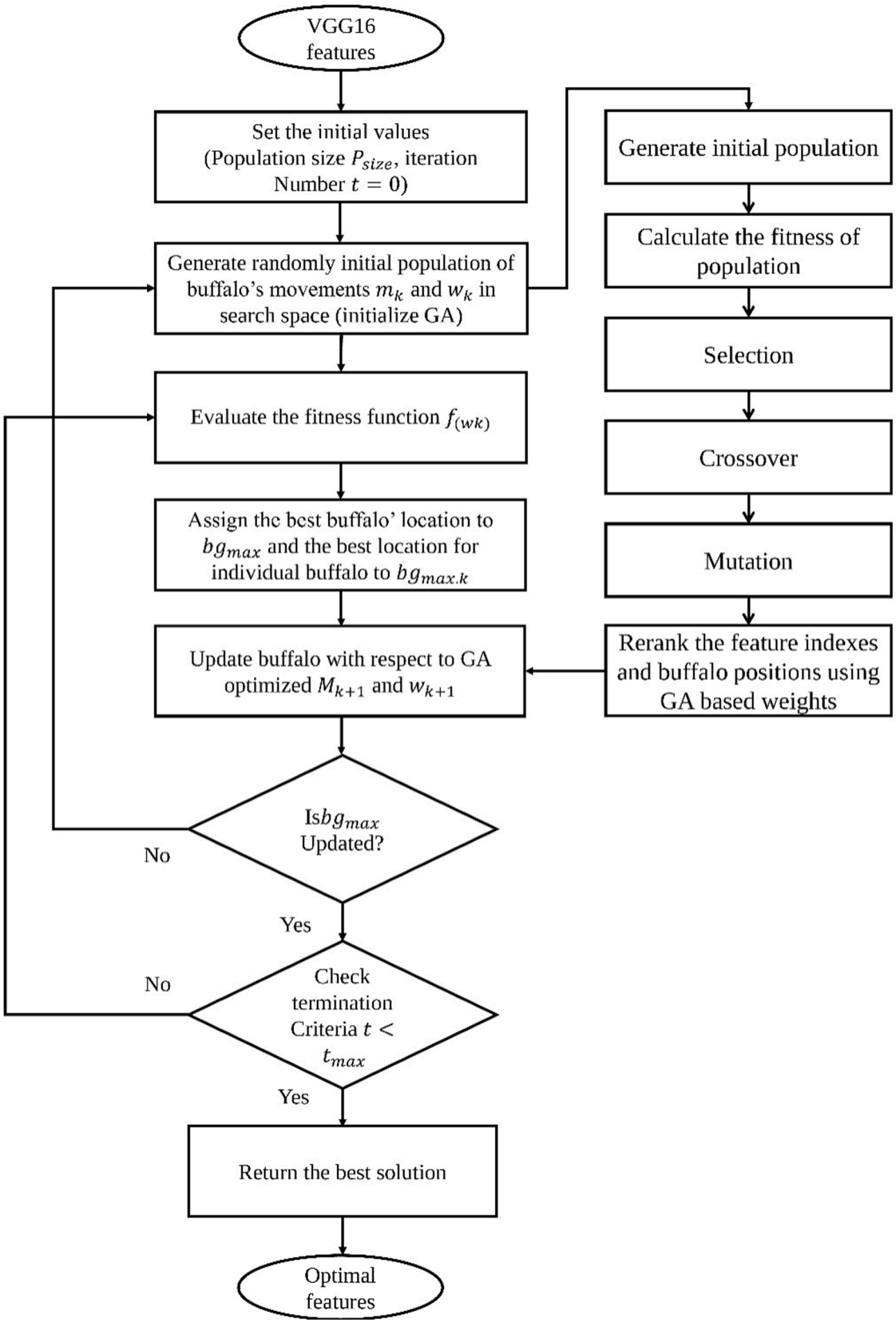


Fig. 3. HABO-GA feature selection process.

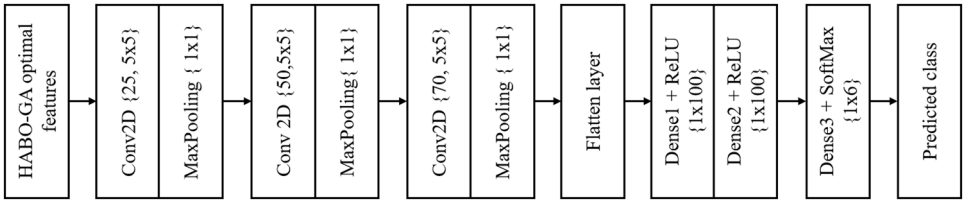


Fig. 4. LC-CNN model.

preprocessing is performed to normalize the gene-expression data by replacing the unknown characters and missing symbols. In practical terms, this preprocessing is mainly performed to reduce the class balance issues. Then, VGG16 model is applied on pre-processed dataset, which extracts the correlated disease features through the data analysis. Here, VGG16 model contains multiple number of fixed layers, which are used to identify the best features. However, the VGG16 features are more in quantity, which needs to be reduced through feature selection process. So, HABO-GA is used to select the optimal features through natural inspired hunting (searching) properties. Finally, LC-CNN model is trained with HABO-GA features, which also performs the multi-class cancer classification from test data.

VGG16 architecture

The VGG16 is the advanced transfer learning model, which is loaded with the pretrained weights with multiple number of layers. Fig. 2 shows the layered architecture of VGG16, where green colour boxes indicate the number of repetency of each layer. The VGG16 model is a series of convolutional layers followed by one or a few dense (or fully connected (FC)) layers. From the input layer to the last max pooling layer (labelled by $7 \times 7 \times 512$) is regarded as feature extraction part of the model, while the rest of the network is regarded as feature classification part of the model. Table 1 shows each layer size with number of feature maps, feature size, kernel size, stride and activation. Here, Filters contains the important weights, which have a spatial relationship to each other. All convolutional layers use 3×3 filters, which are small and perhaps easy to interpret. An architectural concern with a deep learning network is that the depth of a filter must match the depth of the input for the filter (e.g., the number of channels). The weight values will likely be small positive and negative values centred around 0.0. The pooling layers are used to select the maximum values and dense layers are classify the class specific features.

HABO-GA

The HABO-GA algorithm can be obtained by integrating the GA operator into ABO algorithm along with the new concept of swap sequence operator *SS*. Given a task T requires a set of skills s_j, $j = \{1,2, \ldots, d\}$ to be achieved where d is the number of required skills. Pool of experts X exists in social network $G = (X, E)$, a given task T associated with the set of required skills to achieve the task; $T = [S_1, S_2, \ldots, S_n]$. Fig. 3 shows the block diagram of proposed HABO-GA process, which selects the best features from VGG16 output features.

Initialization

Randomly initialize the main parameters of HABO-GA such as population size P_{size}, learning parameters $lp1$ and $lp2$ and the maximum number of iterations $t_{\max}$. The HABO-GA sets the movement (exploitation and exploration move) for all buffalos randomly (i.e., w_k and m_k), where each buffalo represents a vector of random skills that form a task and exploitation move is a swap sequence made-up of sequence of random swap operator $\widetilde{SO}$.

Evaluation of buffalos

Each buffalo (i.e., solution) in the population evaluated by calculating its objective function $f(w_k^t)$ and assign the $bg_{\max}^t$ and $bg_{\max.k}^t$.

$$f\left(w_k^t\right) = \sum_{i=1}^{n} \sum_{j=i+1}^{n} e\left(x_i, x_j\right) \tag{1}$$

where n is the number of team members, and $e(x_i, x_j)$ is the sum of communication cost between experts (i.e., x_i *and* x_j) in social network G.

Integration of discrete crossover operator using GA

To guarantee that the whole population moving towards the global optimum solution, we integrate a Discrete Crossover operator that uses a random real number to create one child from two parents as **{Parent1}**, **{Parent** 2}, and**{Child}**.The child can be generated by selecting genes of both the parents uniformly depending on the random real number $\mu \in \langle 0,1 \rangle$. The DC applied between the global herd's best solution $bg_{\max}^t$ and the current solution w_k^t to produce a new solution (i.e., child). The produced solution is denoted as $w_{k-cross}^t$.

Update (exploitation move)

The initial buffalo's exploitation move m_k^t is made-up of random set of swap operators $\widetilde{SO}$ and can be updated according to

$$m_k^{t+1} = GA\left(m_k^t \oplus lp1 \otimes \left(w_{k-cross}^t - w_k^t\right) \oplus lp2 \otimes \left(bg_{\max.k}^t - w_k^t\right)\right) \tag{2}$$

where $lp1$ and $lp2$ are learning parameters randomly between [0,1], $\oplus$ is a combining sequence operator of two swap operators, $\otimes$ is the probability of $lp1$ that all swap operators can be selected from the swap sequence $w_{k-cross}^t - w_k^t$ and the probability of $lp2$ that all swap operators can be selected from the swap sequence $bg_{max} - w_k^t$ which are denoted as SS_1 and SS_2 respectively. Therefore, $m_k^{t+1} = SS = (\widehat{SO}_1, \widehat{SO}_2, \ldots, \widehat{SO}_l)$ where l is the number of swap operators in m_k^t, SS_1, $SS1$ and SS_2. Here, GA is used to optimize these buffalo location-based weight coefficients.

Update (exploration move)

To update the movement of current buffalo, we apply the sequence of swap operators to the current solution to obtain a new solution with a new movement.

$$w_k^{t+1} = \frac{\left(w_k^t + m_k^t\right)}{\lambda} \ \forall k, \ k = \{1, \ldots, P_{size}\} \tag{3}$$

where λ is a unit time interval over the exploration move and set to 1.

bg_{max}^t check

Check if the herd's best fitness value updated or not; if $bg_{\max}^{t+1} > bg_{\max}^t$, then repeat the process from step 2 while the termination condition $t_{\max}$ not satisfied. Otherwise, back to step 1 and repeat the process again.

Output the best solution

After several iterations, the global best solution with minimum communication cost among team members obtained as a final solution.

Table 2
Dataset description.

Dataset	Number of features	Number of samples	HABO-GA features
Lymphography	18	148	103
Ovarian cancer	46	253	110
Leukemia	22,282	635	114
Lung cancer	4	64	110
Prostate cancer	8	354	108

LC-CNN classification

The LC-CNN architecture for classification includes multiple layers, i.e., convolution (Conv2D), max-pooling, and FC layers. Here, convolution layers are meant for feature detection, Max Pooling layers are meant for reduction of features, which resulted in best feature selection. Fig. 4 shows the LC-CNN classification process.

Conv2D layer

The first layer of a LC-CNN is always a convolutional Layer, which performs the convolution operation between input to kernel, passing the result to the next layer. A convolution converts all the pixels in its receptive field into a single value.

Max pooling

Pooling is a feature commonly imbibed into CNN architectures. The main idea behind a pooling layer is to "accumulate" features from maps generated by convolving a filter over a data. Formally, its function is to progressively reduce the spatial size of the representation to reduce the number of parameters and computation in the network. The most common form of pooling is max pooling.

ReLU

ReLU is a non-linear activation function that is used in multi-layer neural networks or deep neural networks. This function can be represented as:

$$f(x) = \max(0, x) \tag{4}$$

Dense layers

The dense layers are used to connect multiple neurons each other by correlation mapping. In dense layers, after the application of the SoftMax function, is to test the reliability of the model using as loss function the cross-entropy function, to maximize the performance of our neural network. Finally, the SoftMax classifier identifies the various diseases from HABO-GA extracted features.

Results and discussion

This section deals with the detailed analysis of dataset description and obtained quantitative results using existing GECC approaches and proposed GeneNet approach.

Dataset

The structure of each dataset is laid out in Table 2, including the number of features that are available, the number of samples, and the number of classes. These datasets are obtained from the library at the University of California, Irvine. In order to eliminate the numerical issue, each dataset has been standardized.

Performance evaluation

This section gives the performance evaluation and comparison of proposed GeneNet on various publicly available datasets such as ovarian, lymphography, leukemia, lung, and prostate cancer datasets. The quality measurements considered to assess the effectiveness of proposed and state-of-the-art GECI approaches are accuracy, precision, recall, F1-score, and computational time. Table 3 listed the obtained quality metrics for ovarian cancer dataset, whereas the proposed GeneNet outperformed the existing GECI approaches with 36.18 % of increment in accuracy, 25.4 %, 32. 27 % of improvement in precision,

Table 3
Performance estimation on ovarian cancer dataset.

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score	Execution time (s)
WGGL [19]	65.251	60.412	62.666	63.414	9.777
TLBOGSA [24]	67.581	70.836	68.685	78.522	8.656
LS-CNN [23]	73.436	79.740	75.602	79.940	5.577
GECC-Net [29]	94.28	95.23	93.75	94.16	**0.019**
Proposed GeneNet	**99.45**	**99.27**	**99.36**	**99.63**	0.88

Table 4
Performance estimation on lymphography dataset.

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score	Execution time (s)
WGGL [19]	64.644	60.931	61.481	65.027	8.692
TLBOGSA [24]	70.991	64.508	66.441	66.441	6.244
LS-CNN [23]	76.873	**78.625**	**74.445**	**74.304**	4.212
GECC-Net [29]	76.66	53.70	55.55	51.95	**0.005**
Proposed GeneNet	**86.66**	67.27	69.77	68.31	0.68

Table 5
Performance estimation on leukemia dataset.

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score	Execution time (s)
WGGL [19]	63.753	66.318	60.273	60.693	6.715
TLBOGSA [24]	71.354	70.370	70.187	66.625	3.873
LS-CNN [23]	72.443	**71.558**	**71.254**	**68.56**	2.293
Proposed GeneNet	**76.92**	71.42	65.99	65.96	**0.746**

Table 6
Performance estimation on lung cancer dataset.

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score	Execution time (s)
WGGL [19]	64.366	62.740	62.118	60.541	6.570
TLBOGSA [24]	70.740	67.576	67.165	62.770	4.815
LS-CNN [23]	71.840	73.199	73.411	73.084	2.244
Proposed GeneNet	**99.18**	**99.45**	**98.91**	**99.72**	**0.894**

Table 7
Performance estimation on prostate cancer dataset.

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score	Execution time (s)
WGGL [19]	61.227	63.216	61.425	65.579	6.531
TLBOGSA [24]	70.625	67.641	61.997	67.109	4.833
LS-CNN [23]	74.963	76.759	73.296	73.386	2.819
Proposed GeneNet	**99.54**	**99.81**	**99.81**	**99.9**	**0.820**

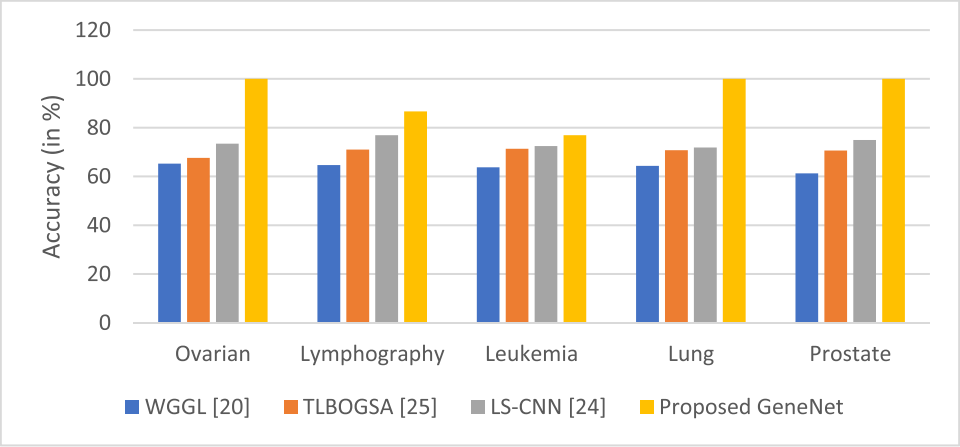


Fig. 5. Performance comparison of accuracy obtained using existing and proposed GeneNet-based GECC approaches.

and recall, respectively.

Similarly, the performance estimation on other cancer datasets such as lymphography, leukemia, lung, and prostate cancer are presented in Tables 4, 5, 6, and 7, respectively. Here, the proposed GeneNet resulted in superior performance for all metrics as compared to state-of-the-art approaches such as WGGL [19], TLBOGSA [24], and LS-CNN [23]. Table 4 listed the obtained quality metrics for lymphography cancer dataset, whereas the proposed GeneNet outperformed the existing GECI approaches with 12.65 % of increment in accuracy, and 83.84 % decrement in computational time, respectively. Table 5 demonstrates the obtained quality metrics for leukemia cancer dataset, whereas the proposed GeneNet outperformed the existing GECI approaches with 6.18 % of increment in accuracy, and 67.68 % decrement in computational time.

Table 6 listed the obtained quality metrics for lung cancer dataset, whereas the proposed GeneNet performed superior to the existing GECI approaches with 39.19 %, 36.63 %, 36.23 %, and 36.83 % of increment in accuracy, precision, recall, and F1-score respectively. It has got 60.26 % decrement in computational time. Finally, Table 7 listed the obtained quality metrics using WGGL, TLBOGSA, and LS-CNN with proposed GeneNet for prostate cancer dataset. In addition, Fig. 5 disclose that the performance comparison of accuracy obtained using existing and proposed GECI approaches.

Conclusion

This work developed a detailed analysis of the proposed GeneNet that is used to classify the cancer disease based on gene expression datasets derived from lymphography, ovarian, leukaemia, lung, and prostate cancers. At first, dataset preprocessing is carried out in order to normalise the gene-expression data. This is accomplished by replacing any unknown characters or symbols that were omitted. The primary objective of the preprocessing of the dataset was to address class balance concerns. After that, a transfer learning VGG16 model was applied to the pre-processed dataset. This extracted the correlated disease features through the analysis of the data. In this case, the VGG16 model includes a large number of fixed layers. These layers are used to determine which features are the most useful. The VGG16 features, on the other hand, come in a greater quantity, which means that the selection process for those features needs to be refined. Therefore, HABO-GA is utilised to select the optimal properties through the use of properties that are naturally inspired by hunting or searching. At the end of the process, an LC-CNN model is trained with HABO-GA features. This model also performs the multi-class cancer classification based on test data. Further, this work can be extended with other optimization models and transfer learning models for improved performance.

Research involving human participants and/or animals

This article does not contain any studies involving animals performed by any of the authors.

Informed consent

Not applicable.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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